

Xiaoxiao Lei

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EMPLOYMENT

J.P. Morgan Chase & Co.

Dallas, Tx, United States

Associate Software Engineer

Apr 2022 – Present

- Led a **Google ADK**-based financial-instrument reconciliation **agent system** that detects discrepancies from data from vendor emails against platform persisted records. The system combined instrument-type routing, text extraction, and sequential/parallel comparison in a schema-guided multi-agent workflow using **FastAPI** and **AWS Bedrock**. In manually reviewed test emails, the prototype identified **95%** of known discrepancies and was projected to save approximately **1.5** hours of daily manual review per banker.
- Built a **Databricks lakehouse** for capital-markets reference data following a **medallion architecture**. Managed schema definitions and changes with **Liquibase**, and provisioned Databricks compute, SQL warehouses, Unity Catalog resources, and storage through **Terraform**. Developed **PySpark**, **Delta Live Tables**, **Auto Loader**, and **AWS Glue** pipelines to ingest Bloomberg financial-data feeds, normalize and validate records, maintain historical versions, and publish curated datasets for downstream consumers, improving data consistency, traceability, and onboarding of new datasets.
- Co-built an end-to-end **AWS data pipeline** for a financial-instrument platform, ingesting **40,000+** offering records per day from third-party vendor APIs and transforming **200+** attributes using **PySpark on AWS Glue**. Normalized vendor records into validated domain objects under a consistent data model, persisted canonical records to **Aurora PostgreSQL**, and published enriched events through **SNS** for downstream applications.
- Developed a suite of **Spring Boot microservices** on **ECS** for financial-data enrichment and document management across the Market Monitor platform. Built **GraphQL/REST APIs** that enabled users to preview, correct, query, and consume normalized financial records, and supported document search, upload, and download through a **MongoDB**-backed broker service. Implemented extensible **RBAC permission resolution** based on user roles, document types, and request attributes, with audit logging for traceability. Translated evolving business requirements into specifications and leveraged **GitHub Copilot** to accelerate development. **Automated** integration tests using **AI agent skills** and scripts
- Strengthened **production reliability** and operational readiness by building an **AWS Lambda** utility that injected **30+** record types on demand for end-to-end testing, reducing debugging time by **80%**; provided production support by monitoring and triaging alerts, tracing cross-service failures with **Datadog** and **AWS CloudWatch**, performing root-cause analysis, and coordinating prioritized remediation work through resolution.
- Owned **infrastructure delivery** for an AI agent governance platform that validates architecture proposals against configurable rules. Built a repeatable, multi-environment AWS delivery foundation by provisioning cloud resources through **Terraform IaC**, deploying containerized services to **ECS clusters**, and manage **CI/CD pipelines** across environments with **Jenkins**, **Jib**, and **Spinnaker**.
- Delivered end-to-end development of a **React**-based admin portal for an internal fundraising platform, translating **Figma** designs and complex authorization rules into intuitive permission-management workflows; built reusable UI components within a monorepo and integrated the front end with **GraphQL/REST APIs** for managing read/write access to financial and entity records.

EDUCATION

State University of New York at Buffalo

Buffalo, NY, USA

Master of Science in Computer Science and Engineering

Jan 2020 – Feb 2022

SKILLS

Certifications: [AWS Certified Developer - Associate](#), [Apollo Graph Developer - Professional](#)

Generative AI & Agents: Google Agent Development Kit (ADK), AWS Bedrock, agent orchestration, prompt engineering; Github Copilot, Claude Code, Codex, Spec Driven Development, LangChain, LangGraph, RAG, Vector DataBase, StreamLit

Languages: Java, Python, SQL, JavaScript, TypeScript, HTML, CSS

Frameworks: Spring Boot, FastAPI, Flask, Node.js, React, Next.js, GraphQL, RESTful APIs, Kafka, RBAC, Spring Security, OAuth, JWT, Playwright, Cucumber

Database: MySQL, PostgreSQL, NoSQL, MongoDB, Liquibase, Databricks

Cloud DevOps: AWS EC2, S3, ECS, EKS, Lambda, Glue, SNS/SQS, Aurora, DynamoDB, CloudWatch, Terraform IaC, GCP, Docker, Kubernetes, Jenkins, Spinnaker

Observability: Datadog, Splunk, Grafana, Prometheus